Exercise Sheet 10: Hypothesis Testing

One-sample procedures (z / t / chi-square, proportions) + selected two-sample and chi-square methods

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All expected counts are above 5. 11

Quick reference: hypothesis testing building blocks

Core objects

- **Null hypothesis** H_0: baseline claim about a parameter (e.g. $\mu = \mu_0$).
- **Alternative** H_1: departures from H_0 we want to detect (two-sided or one-sided).
- **Test statistic** W: a function of the sample data.
- **Rejection region** R: reject H_0 iff $W_{\text{obs}} \in R$.
- **Type I error**: reject H_0 although H_0 is true. Probability α.
- **Type II error**: fail to reject H_0 although H_1 is true. Probability β.
- **Power**: $1 - \beta$.

Quantiles and p-values in R

- Normal critical value: `qnorm(1 - alpha/2)` and CDF: `pnorm()`
- t critical value: `qt(1 - alpha/2, df)` and CDF: `pt()`
- chi-square quantile: `qchisq(p, df)` and CDF: `pchisq()`

Common tests in R (and what they correspond to)

- One-sample mean (unknown σ): `t.test(x, mu = mu0)`
- One-sample mean (known σ): compute Z and use `pnorm()`
- One-sample variance (normal model): compute Q and use `pchisq()`
- One-sample proportion (approx): `prop.test(x, n, p = p0, correct = FALSE)`
- One-sample proportion (exact): `binom.test(x, n, p = p0)`
- Two-sample means (Welch): `t.test(x, y)` (default is Welch)
- Two-sample means (pooled, equal variances): `t.test(x, y, var.equal = TRUE)`
- F test for equal variances (normal): `var.test(x, y)`
- Chi-square test of independence (contingency table): `chisq.test(tab)`
- Wilcoxon–Mann–Whitney (rank-based): `wilcox.test(x, y)`
- Paired t-test: `t.test(x, y, paired = TRUE)`

Part A — Logic of hypothesis testing (conceptual)

Exercise 1 (conceptual): H_0, H_1, statistic, rejection region

We use `mtcars$mpg` as a sample.

```r
data(mtcars)
x <- mtcars$mpg
n <- length(x)
c(n = n, mean = mean(x), sd = sd(x))
```

```
##         n      mean        sd
## 32.000000 20.090625  6.026948
```

Tasks

1. Write down H_0 and H_1 for each case (in symbols and one short sentence):

   - (a) Two-sided: "the mean mpg equals 20". H0: The mean mpg equals 20 H1: The mean mpg does not equal 20
   - (b) Right-tailed: "the mean mpg is larger than 20". H0: The mean mpg is smaller or equal to 20 H1: The mean mpg is larger than 20
   - (c) Left-tailed: "the mean mpg is smaller than 20". H0: The mean mpg is larger or equal to 20 H1: The mean mpg is smaller than 20

2. For the one-sample t-test, the test statistic is

$$T = \frac{\bar{X} - \mu_0}{S/\sqrt{n}}.$$

   Explain (in 2–3 sentences) what "large" values of T mean in words.

   A: A large value for T indicates that he sample mean X is far away from the actual mean, arguing against H0.

3. What is a **rejection region** R for a two-sided test at level $\alpha = 0.05$? (Answer using a critical value $t_{n-1,0.975}$.)

$$R = \{T : T < -t_{n-1,0.975} \text{ or } T > t_{n-1,0.975}\}$$

4. In 1–2 sentences each: define Type I error, Type II error, and power.

   Type I Error: Falsely rejecting H_0

   Type II Error: Failing to reject H_0 when it should be.

   Power: Probability of correctly rejecting H_0

Part B — One-sample tests in the normal model

Exercise 2: One-sample t-test for a mean (unknown σ) + p-value

We test the mean mpg in `mtcars`.

Let $\mu_0 = 20$ and $\alpha = 0.05$.

Tasks

1. Compute $\bar{x}$, s, n, and the observed test statistic

$$t_{\text{obs}} = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}.$$

```
meanx <- mean(mtcars$mpg)
s <- sd(mtcars$mpg)
n <- length(mtcars$mpg)
```

2. Compute the **two-sided p-value** manually using the t-distribution:

$$p = 2\,P\left(T \geq |t_{\text{obs}}|\right), \quad T \sim t_{n-1}.$$

```r
t_obs <- (meanx - 20)/(s / sqrt(n))
p_2sided <- 2 * (1 - pt(abs(t_obs), df = n-1))
```

3. Check with `t.test()` and confirm that the p-value matches.

```r
t.test(mtcars$mpg)
```

```
##
##  One Sample t-test
##
## data:  mtcars$mpg
## t = 18.857, df = 31, p-value < 2.2e-16
## alternative hypothesis: true mean is not equal to 0
## 95 percent confidence interval:
##  17.91768 22.26357
## sample estimates:
## mean of x
##  20.09062
```

4. State your decision at $\alpha = 0.05$: reject or not reject H_0? (Write 1–2 sentences: include p-value and conclusion in context.)

   The p-value is ~0.9328, which is much larger then 0.05. $\rightarrow$ H_0 cannot be rejected. $\rightarrow$ There is no signifitant evidence that the mean of mpg differs from 20.

Exercise 3: CI–test duality for the mean

Use the same data and $H_0 : \mu = \mu_0$ with $\mu_0 = 20$ and $\alpha = 0.05$.

Tasks

1. Compute the **95% confidence interval** for μ using `t.test()`.

```r
ttest <-t.test(mtcars$mpg)
ttest$conf.int[1:2]
```

```
## [1] 17.91768 22.26357
```

2. Verify CI–test duality: check whether μ_0 lies inside the CI. Explain why this agrees with your test decision in Exercise 2.

   μ_0 lies inside the CI. This is because both the calculation for the CI aswell as the calculation for μ_0 use the same logic / formula, just isolated on another value. Because you can also derive from $\mu_0 \in$ CI that we cannot drop H_0, the same result as the t-test. This shows the duality of both approaches.

3. Repeat (conceptually only): what happens to the decision rule if you change the confidence level from 95% to 99%?

   The CI grows.

Exercise 4: One-sample z-test for a mean (known σ) (illustration)

In practice σ is rarely known. Here we do an **illustration** using `airquality$Ozone`.

Assume (for this exercise) that the population standard deviation is known and equals $\sigma_0 = 20$. We test $H_0 : \mu = 40$ vs $H_1 : \mu \neq 40$ at $\alpha = 0.05$.

```
x <- airquality$Ozone
sigma <- 20
mu_0 <- 40
alpha <- .05
```

Tasks

1. Compute the z-statistic

$$z_{\text{obs}} = \frac{\bar{x} - \mu_0}{\sigma_0/\sqrt{n}}$$

   with $\mu_0 = 40$ and $\sigma_0 = 20$.

```
z_obs <- (mean(x, na.rm=T) - mu_0) / (sigma / sqrt(length(x)))
z_obs
```

```
## [1] 1.316906
```

2. Compute the two-sided p-value using the standard normal distribution.

```
p_2sided <- 2 * (1 - pnorm(abs(z_obs)))
p_2sided
```

```
## [1] 0.1878702
```

3. Compute the critical value $z_{0.975}$ and decide whether to reject.

```
z_crit <- qnorm(1 - alpha / 2)
z_crit
```

```
## [1] 1.959964
```

```
#-> I do not reject H_0
```

4. In 1–2 sentences: why is the z-test "more optimistic" than the t-test when n is small?

   Because the z test assumes normal distrobution, allowing it to assume a bigger interval, as it therefore assumes that the standard deviation is known. This underestimated the true variability of the given distribution.

Exercise 5: One-sample chi-square test for a variance

Assume that `mtcars$mpg` is approximately normal. We test

$$H_0 : \sigma^2 = \sigma_0^2 \quad \text{with } \sigma_0^2 = 36$$

(two-sided, $\alpha = 0.05$).

Under H_0,

$$Q = \frac{(n-1)S^2}{\sigma_0^2} \sim \chi^2_{n-1}.$$

Tasks

1. Compute S^2, Q_{obs}, and the two-sided p-value.

```
S_squared <- sd(mtcars$mpg, na.rm=T)^2
Q <- 31 * S_squared / 36
p_2sided <- 2 * min(
  pchisq(Q, df=31),
  1 - pchisq(Q, df=31)
)
```

```
p_2sided
```

```
## [1] 0.9044279
```

2. Decide at $\alpha = 0.05$. (Reject if in either tail.)

```
p_2sided <= .05
```

```
## [1] FALSE
```

The p-value is bigger then alpha → don't reject H_0

3. State H_0 in words, and interpret the result in 1–2 sentences.

H_0: $\sigma_0^2 = 36$

Result: There is not enough evidence to reject H_0. This is the case, because the calculated p-value is much bigger the given alpha, resulting in failure of disproving H_0

Part C — One-sample tests for a proportion (Bernoulli/binomial)

Exercise 6: Proportion test (normal approximation vs exact)

We use the built-in table `Titanic` and focus on the overall survival proportion.

Let n = total passengers, x = number of survivors, $\hat{p} = x/n$. Test $H_0 : p = 0.38$ vs $H_1 : p > 0.38$ at $\alpha = 0.05$.

```r
data(Titanic)
tab <- Titanic
n <- sum(tab)
x <- sum(tab[,,, "Yes"])
phat <- x / n
c(n = n, x = x, phat = phat)
```

```
##           n           x        phat
## 2201.000000  711.000000    0.323035
```

Tasks

1. Compute the normal-approximation z-statistic under H_0:

$$Z = \frac{\hat{p} - p_0}{\sqrt{p_0(1 - p_0)/n}}.$$

```r
Z <- (phat - .38) / (sqrt(.38 * (1 - .38) / n))
```

2. Compute the one-sided p-value $p = P(Z \geq z_{\text{obs}})$ and decide at $\alpha = 0.05$.

```r
p_1sided <- 1 - pnorm(Z)
p_1sided
```

```
## [1] 1
```

Because the p-value is much bigger then alpha: $1 > 0.05$, H_1 is not support and H_0 can definitely not be rejected.

3. Compute the **exact** binomial p-value using `binom.test()` (one-sided, alternative = "greater"). Compare the two p-values and comment briefly.

```r
binom.test(x, n, .38, alternative = "greater")$p.value
```

```
## [1] 1
```

identical.

4. Check the rule of thumb for the normal approximation: are np_0 and $n(1 - p_0)$ at least 10?

```r
n * .38 > 10
```

```
## [1] TRUE
```

```r
n * (1 - .38) > 10
```

```
## [1] TRUE
```

$\rightarrow$ yes

Exercise 7: p-values for two-sided vs one-sided alternatives

Use your t_{obs} from Exercise 2.

Tasks

1.

```
t_obs
```

```
## [1] 0.08506004
```

(a) two-sided $H_1 : \mu \neq \mu_0$,

(b) right-tailed $H_1 : \mu > \mu_0$,

(c) left-tailed $H_1 : \mu < \mu_0$.

```
#a
2 * (1 - pt(abs(t_obs), df = 31))
```

```
## [1] 0.9327606
```

```
#b
1 - pt(t_obs, df=31)
```

```
## [1] 0.4663803
```

```
#c
pt(t_obs, df=31)
```

```
## [1] 0.5336197
```

2. Explain in 2–3 sentences why one-sided p-values are (typically) about half of two-sided p-values when t_{obs} is in the direction of H_1.

The closer t_{obs} is to H_0, the more equal both one-sided p-values become, as the bigger one is getting smaller and the smaller one is getting bigger (trivial). Because the 2-sided p-value is the p-value in both directions, it's getting closer to left-sided-p + right-sided-p, which, if both are ~ equal, is ~ equal to $2 *$ *-sided-p.

Part D — Type I/II error and power (simulation)

Exercise 8: Power vs significance level (one-sample t-test)

We simulate a one-sample t-test for $H_0 : \mu = \mu_0$ (two-sided). Assume the data are normal with $\sigma = 4$, $n = 25$, and $\mu_0 = 20$.

We estimate: - α (Type I error) when the true mean is $\mu = 20$, - power when the true mean is $\mu = 22$.

```r
sigma <- 4
n <- 25
mu_0 <- 20
```

Tasks

1. Interpret the four numbers:
   - Why is $\widehat{\alpha}$ close to the chosen α?
   - Why does power drop when you lower α from 0.05 to 0.01?

-

2. (Conceptual) Explain why a one-sided test can have higher power than a two-sided test at the same α. When is a one-sided test not justified?

-

Part E — Selected two-sample and chi-square methods (extension)

Exercise 9: Two-sample means (pooled vs Welch) using `sleep`

The dataset `sleep` contains increases in sleep for two drugs.

```r
data(sleep)
head(sleep)
```

```
##   extra group ID
## 1   0.7     1  1
## 2  -1.6     1  2
## 3  -0.2     1  3
## 4  -1.2     1  4
## 5  -0.1     1  5
## 6   3.4     1  6
```

Let X be group 1 and Y group 2 (`extra` by `group`).

```r
x <- sleep$extra[sleep$group == 1]
y <- sleep$extra[sleep$group == 2]
c(n1 = length(x), n2 = length(y), mean1 = mean(x), mean2 = mean(y), sd1 = sd(x), sd2 = sd(y))
```

```
##        n1        n2     mean1     mean2       sd1       sd2
## 10.000000 10.000000  0.750000  2.330000  1.789010  2.002249
```

Tasks

1. Perform Welch's two-sample test (default) and report the p-value.
2. Perform the pooled two-sample t-test (`var.equal=TRUE`) and report the p-value.
3. In 2–3 sentences: when would you prefer Welch over pooled?

```r
t.test(x, y, alternative = "two.sided", var.equal = FALSE)$p.value
```

```
## [1] 0.07939414
```

```r
t.test(x, y, alternative = "two.sided", var.equal = TRUE)$p.value
```

```
## [1] 0.07918671
```

I prefer the Welch test, as the pooled test needs very precise preconditions, like almost identical variance and n with reason (which is not given)

Exercise 10: Chi-square test of independence using `Titanic`

Test whether survival is independent of sex (ignoring other dimensions).

```r
tab2 <- apply(Titanic, c("Sex", "Survived"), sum)
tab2
```

```
##         Survived
## Sex        No Yes
##   Male   1364 367
##   Female  126 344
```

Tasks

1. State H_0 in words.
2. Compute the chi-square test and interpret the p-value.
3. Check whether expected counts are all at least 5.

H_0: Survival is independent of sex.

```r
ct <- chisq.test(tab2)
ct
```

```
##
##  Pearson's Chi-squared test with Yates' continuity correction
##
## data:  tab2
## X-squared = 454.5, df = 1, p-value < 2.2e-16
```

p-value $= 2.2e - 16$

$\rightarrow$ The p-value is essentially zero, providing overwhelming evidence against the null hypothesis. We reject H_0 and conclude that survival is not independent of sex.

```r
ct$expected
```

```
##         Survived
## Sex             No      Yes
##   Male   1171.8264 559.1736
##   Female  318.1736 151.8264
```

All expected counts are above 5.